



Figure 7: Embedded Linux runs on PowerPC 405 from IBM

MHz (see Figure 7). ASM provides the same controls as the serial-line SMS but in a more usable manner (Figure 8). It lets you manage the system remotely—start/stop it, view logs, allocate resources (like *Capacity on Demand* parameters), etc.

Let's briefly see how this subsystem is launched.

When you connect the mains power to the box, the ServiceProcessor automatically starts the embedded Linux. After initially testing some components of a larger block, ServiceProcessor displays on a front panel the readiness status for full system start. When you hit the 'Power On' button, OpenFirmware loads and ultimately passes control to your Linux boot loader, *yaboot*.

Like with the large system, this small subsystem has two independent Ethernet controllers, referred to as *HMC 1* and *HMC 2*. It makes sense to connect them to separate LANs/VLANs for a special management network. First, it helps access restriction and security. Second, you can avoid a network traffic overload on a single LAN. IP addresses for these ports can be supplied by a DHCP server on a management network. If one isn't found, default addresses are automatically assigned: 192.168.2.147 (port C7-T1, on *HMC1*) and 192.168.3.147 (port C7-T2, *HMC2*). You can log in via *telnet*, with the default credentials *admin/admin*.

```
$ mount
/dev/mtdblock/0/3 on / type cramfs (rw)
none on /dev type devfs (rw)
proc on /proc type proc (rw)
/dev/mtdblock/0/1 on /opt/extcode type jffs2 (rw)
/dev/mtdblock/0/1 on /etc type jffs2 (rw)
/dev/mtdblock/tmpmram on /tmp type jffs2 (rw)
none on /var type tmpfs (rw)
/dev/mtdblock/pcscore/pcscore on /core type jffs2 (rw)
/dev/mtdblock/pcscram/pcscram on /opt/p1 type jffs2 (rw)
shm on /var/shm type shm (rw)
/dev/mtdblock/1/1 on /alt/extcode type jffs2 (rw)
/dev/mtdblock/2/3 on /opt/fips type cramfs (rw)
```



Figure 8: ASM's user-friendly controls presentation

```
/dev/mtdblock/nvram/nvram on /opt/p2 type zofs (rw, sync)
$ uptime
8:40am up 1 day, 14:53, load average: 0.52, 0.20, 0.07
$ uname -a
Linux fsp 2.4.18-rc4 #1 Wed Mar 25 15:01:22 UTC 2009 ppc unknown
$ cat /proc/version
Linux version 2.4.18-rc4 (root@mcbuild6) (gcc version 3.2.3
20030401 (prerelease)) #1 Wed Mar 25 15:01:22 UTC 2009
```

As we see, Embedded Linux uses JFFS/CRAMFS filesystems, which are common in tiny products. An interesting feature is NVRAM memory mounted as storage (*/dev/mtdblock/nvram/nvram*). Reading data from its files gives the status of different subsystems. You can also change some values using commands from */opt/fips/bin*—particularly, you can even try to launch several subsystems (or maybe the full system box) with */opt/fips/bin/powr_fsp_spcn*, which checks the current system status, and accordingly launches */bin/powr_L4* or */bin/powr_sqh*. The ASM Web server (named *webs*), when you click the button *Save settings and power on* does the same.

Available distros for Power

Originally, I planned to use Red Hat Enterprise Linux as the main OS on this system. However, neither RHEL 5.7 nor 6.2 Beta, nor even 6.2 started correctly. RHEL 6.2 Beta and a more recent version 6.2 spews a kernel trap error during initial start-up. Yes, the IntelliStation 285 wasn't in the supported hardware list, but neither was it for SLES 10/11. Still, SLES 10 and 11 support it unofficially. Let us see how the installation process goes, and whether we can bring the system to production mode.

In addition to RHEL and SLES, numerous systems based on the POWER processor even supported Ubuntu 10.04 (LTS edition). However, my experience proved that this